

YOUR COMPONENTS FOR 2-LANE SYSTEM

Component	Quantity	Used For
ESP32 MODULE 38 PIN	2	Traffic light controller + Emergency vehicle
NEO6M GPS	2	Location tracking for both units
GL12 BREADBOARD 840 PT	1	Traffic light circuit
MB-102 BREADBOARD	1	Emergency vehicle circuit
ESP32-38 PIN EXPANSION	1	For vehicle's battery power
2200 mAh L-ion BATTERY	1	Powers the vehicle
18650 1 CELL HOLDER	1	Holds battery safely
TP4056 B TYPE	1	Battery charger/protector
JUMPERS M-M (40 PCS)	1	ESP32 to breadboard connections
JUMPERS F-F (40 PCS)	1	GPS module connections
Push Button	1	Trigger preemption on vehicle
220-ohm Resistors	12	For LEDs (you'll use 6)
LEDs (5mm)	12	You'll use 6 (2 red, 2 yellow, 2 green)
Micro-USB Cables	2	Programming both ESP32s

PART 1: 2-LANE TRAFFIC LIGHT (ESP32 #1 on GL12 Breadboard)

Understanding Your 2-Lane Intersection

For a 2-lane system, you have:

- Lane 1 (Main Street) : Red, Yellow, Green LEDs
- Lane 2 (Cross Street) : Red, Yellow, Green LEDs

Step 1: Place ESP32 on GL12 Breadboard

1. Mount your ESP32 38 PIN on the GL12 breadboard
2. Position it so it straddles the center gap
3. Each pin should be in a separate row for easy access

Step 2: LED Connections (6 LEDs Total)

IMPORTANT RULE: For EVERY LED, connect a 220Ω resistor in series with the positive leg!

Connection pattern for each LED:

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ESP32 GPIO → [M-F Jumper] → 220Ω Resistor → LED Long Leg (+)

ESP32 GND → [M-F Jumper] → LED Short Leg (-)

 2-LANE LED Pin Assignment Table

Lane	LED Color	ESP32 GPIO Pin	Resistor
Lane 1 (Main St)	Red	GPIO 26	220Ω
	Yellow	GPIO 27	220Ω
	Green	GPIO 14	220Ω
Lane 2 (Cross St)	Red	GPIO 25	220Ω
	Yellow	GPIO 33	220Ω
	Green	GPIO 32	220Ω

Step-by-Step: Connecting Lane 1 Red LED

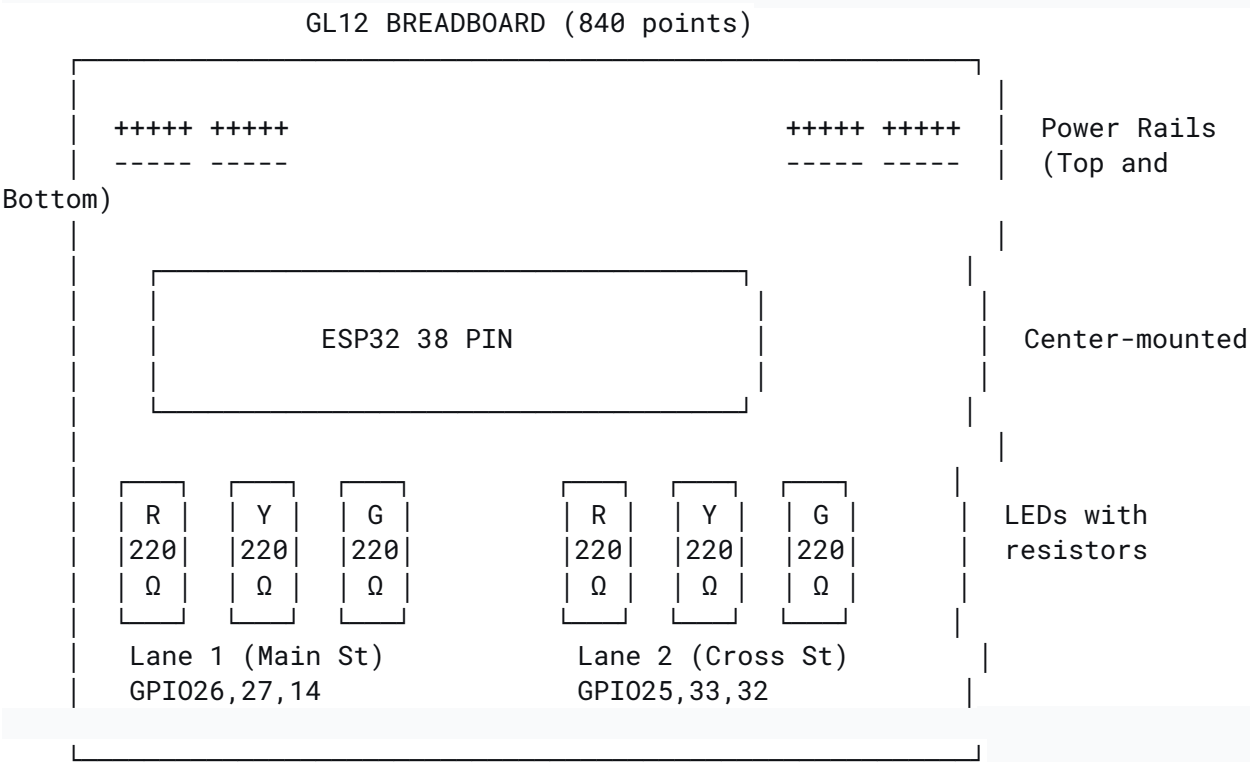
Let me walk you through one complete LED connection:

- 1. Take M-F Jumper wire - Male end goes into ESP32 GPIO 26
- 2. Female end connects to one leg of 220Ω resistor
- 3. Other resistor leg connects to LED Long Leg (+) (longer leg)
- 4. LED Short Leg (-) (shorter leg) connects to another M-F Jumper
- 5. Other end of this jumper connects to any ESP32 GND pin (there are multiple GND pins)

Repeat this exact pattern for all 6 LEDs!

Step 3: Visual Layout on GL12 Breadboard

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Step 4: GPS Connection for Traffic Light ESP32

Using F-F Jumper wires (GPS has male pins):

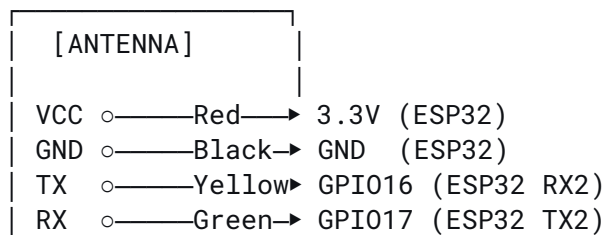
GPS Module Pin	Connect to ESP32 Pin	Wire Color
VCC	3.3V pin	Red F-F
GND	GND pin	Black F-F

TX	GPIO 16 (RX2)	Yellow F-F
RX	GPIO 17 (TX2)	Green F-F

GPS Connection Diagram:

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NEO6M GPS Module



Important GPS Notes:

- GPS needs 3.3V, NOT 5V! (will destroy GPS if you use 5V)
- Default baud rate is 9600
- Place antenna facing up, away from other components
- LED on GPS blinks when satellites found (first fix: 5-15 minutes)

PART 2: EMERGENCY VEHICLE (ESP32 #2 on MB-102 Breadboard + Expansion Board)

Step 1: Mount ESP32 on Expansion Board

1. Align ESP32 module with expansion board socket
2. Antenna should face outward
3. Press firmly but gently until fully seated
4. IMPORTANT: Set the power jumper to 3.3V (not 5V!)

Warning: The pins on the expansion board are offset from the ESP32 pins! Always follow the labels on the expansion board, NOT the ESP32 labels.

Step 2: Place Expansion Board on MB-102 Breadboard

1. Place the expansion board on the MB-102 breadboard
2. It will cover multiple rows - that's fine
3. Note which GPIOs are available on the breadboard rows

Step 3: GPS Connection for Vehicle (Same as Above)

Using F-F Jumper wires:

GPS Module Pin	ESP32 Pin	Wire Color
VCC	3.3V	Red F-F
GND	GND	Black F-F
TX	GPIO 16 (RX2)	Yellow F-F
RX	GPIO 17 (TX2)	Green F-F

Step 4: Push Button Connection

Using M-F Jumper wires:

Button Leg	Connection
One leg	Connect to GPIO 4 using M-F jumper
Other leg	Connect to GND using M-F jumper

Button Wiring Diagram:

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Push Button

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M-F Jumper

M-F Jumper

GPIO 4

GND

No external resistor needed - we'll use ESP32's internal pull-up in code!

PART 3: BATTERY POWER SYSTEM (For Vehicle)

Step 1: Understanding TP4056 B TYPE Terminals

Your TP4056 B TYPE has these terminals:

- B+, B-: Battery connection
- OUT+, OUT-: Output to ESP32
- IN+, IN-: Charging input (Micro-USB port)

 **CRITICAL:** The "B TYPE" may or may not have over-discharge protection. Check if it has a second IC (DW01A) next to the main chip. If not, be careful not to drain battery below 3.0V!

Step 2: Battery Holder Connections

18650 Holder Wire	Connect to TP4056
Red (+)	B+ terminal
Black (-)	B- terminal

Step 3: Insert 2200mAh Battery

- Insert 2200 mAh L-ion BATTERY into holder
- DOUBLE-CHECK polarity! Match + and - markings
- Reversed polarity will destroy the TP4056 instantly!

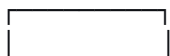
Step 4: TP4056 to ESP32 Expansion Board

TP4056 Terminal	Connect to ESP32 Expansion Board
OUT+	VIN pin (or 5V pin)
OUT-	GND pin

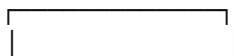
Power System Diagram:

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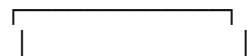
18650 Holder

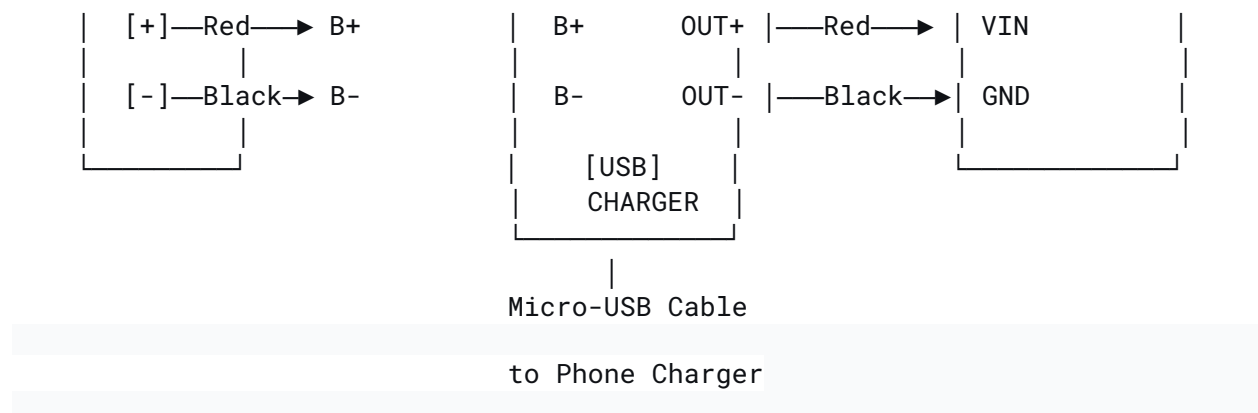


TP4056 B TYPE



ESP32 Expansion





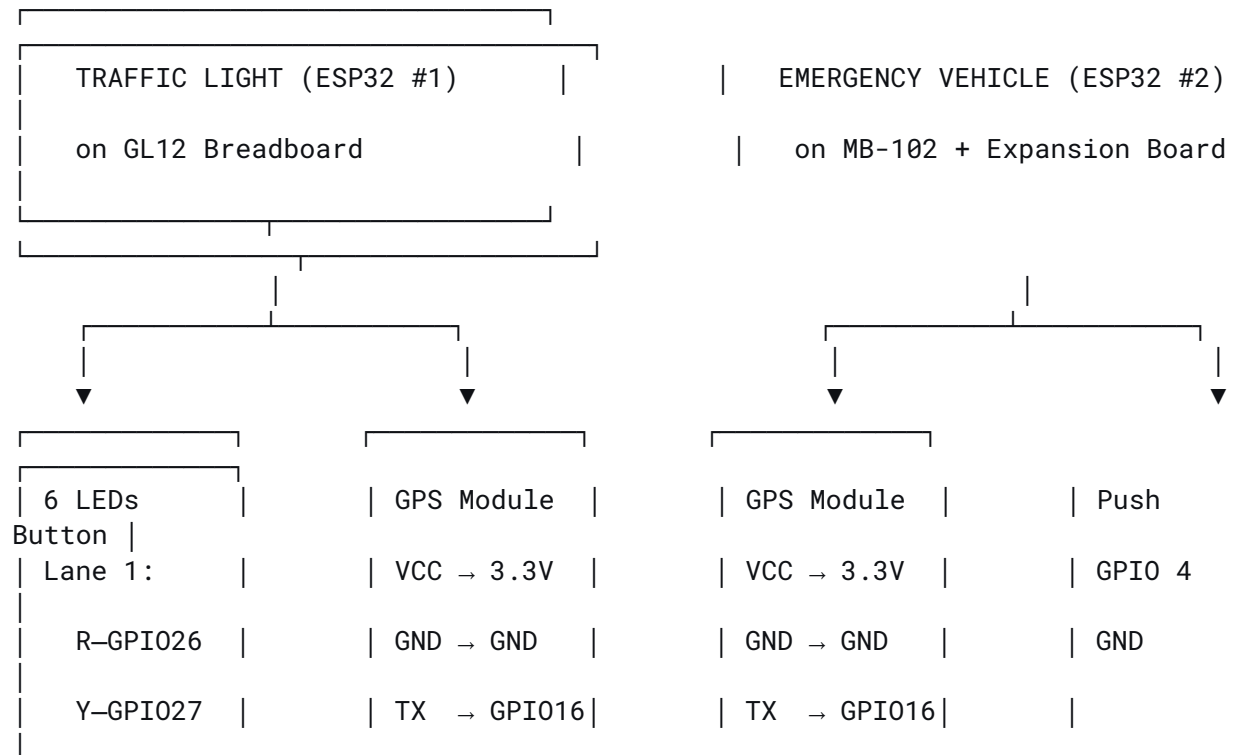
Step 5: Charging the Battery

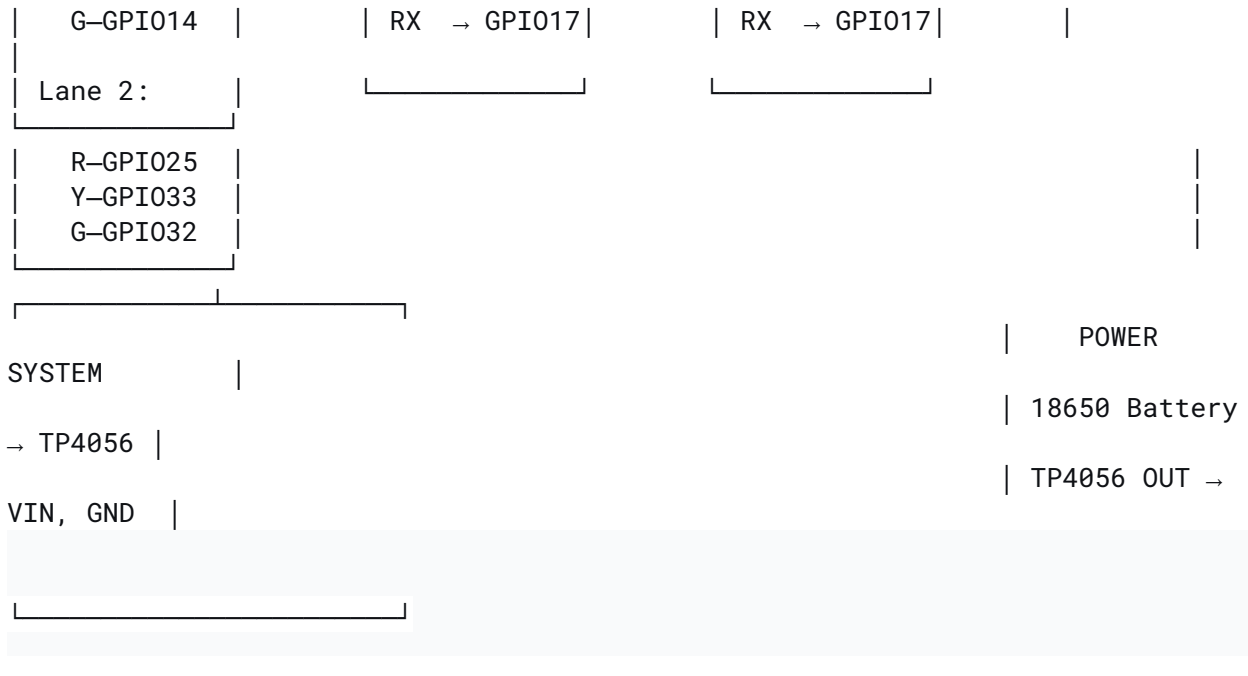
To charge your battery:

- Connect Micro-USB cable to TP4056's charging port
- Red LED = charging
- Blue/Green LED = fully charged
- Charging time for 2200mAh: ~3-4 hours

🌐 PART 4: COMPLETE 2-LANE SYSTEM OVERVIEW

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✓ FINAL CONNECTION CHECKLIST

Traffic Light ESP32 (#1) - GL12 Breadboard:

- ESP32 properly mounted straddling center gap
- Lane 1 Red: GPIO26 → 220Ω → LED+ (long leg), LED- (short leg) → GND
- Lane 1 Yellow: GPIO27 → 220Ω → LED+ (long leg), LED- (short leg) → GND
- Lane 1 Green: GPIO14 → 220Ω → LED+ (long leg), LED- (short leg) → GND
- Lane 2 Red: GPIO25 → 220Ω → LED+ (long leg), LED- (short leg) → GND
- Lane 2 Yellow: GPIO33 → 220Ω → LED+ (long leg), LED- (short leg) → GND
- Lane 2 Green: GPIO32 → 220Ω → LED+ (long leg), LED- (short leg) → GND
- GPS connected: VCC→3.3V, GND→GND, TX→GPIO16, RX→GPIO17 (using F-F jumpers)
- No connections to pins 0,1,3,6-11
- Micro-USB cable ready for power/programming

Vehicle ESP32 (#2) - MB-102 + Expansion Board:

- ESP32 properly seated in expansion board
- Expansion board jumper set to 3.3V (CRITICAL!)
- GPS connected: VCC→3.3V, GND→GND, TX→GPIO16, RX→GPIO17 (F-F jumpers)
- Push button connected: GPIO4 and GND (M-F jumpers)
- Battery inserted in holder with correct polarity (+ and -)
- Holder Red (+) → TP4056 B+
- Holder Black (-) → TP4056 B-
- TP4056 OUT+ → Expansion Board VIN
- TP4056 OUT- → Expansion Board GND
- All connections secure, no loose wires

FIRST POWER-UP SEQUENCE

Step 1: Program Both ESP32s

1. Connect Traffic Light ESP32 to computer via Micro-USB
2. Upload code
3. Disconnect, then connect Vehicle ESP32
4. Upload code to it

Step 2: Power Traffic Light

1. Connect Traffic Light ESP32 to power (USB charger or computer)
2. LEDs should start cycling (Red → Yellow → Green for Lane 1, then Lane 2)
3. GPS LED will start blinking after finding satellites (5-15 minutes)

Step 3: Power Vehicle

1. Connect battery to TP4056
2. Verify expansion board power LED lights up
3. Vehicle GPS will start searching for satellites

Step 4: Test Preemption

1. Wait for both GPS modules to have satellite lock (blinking LEDs)
2. Press the button on vehicle
3. Traffic light should:
 - Turn current green to yellow
 - Turn all red for 2 seconds
 - Turn green for vehicle's lane
 - Hold green for 10 seconds
 - Return to normal cycle

TROUBLESHOOTING QUICK REFERENCE

Problem	Likely Cause	Solution
LED not lighting	Wrong polarity or missing resistor	LED long leg (+) must go to GPIO via resistor, short leg (-) to GND

Multiple LEDs lighting	Short circuit between pins	Check for stray wires touching adjacent pins
No GPS data	Wrong voltage or wiring	Verify VCC is 3.3V (not 5V!), TX→RX, RX→TX
GPS no fix	Indoor or no clear sky	Place near window, wait 15 minutes, LED should blink
ESP32 not powering from battery	TP4056 protection triggered	Charge battery, check OUT+/- connections
Button not working	Wrong GPIO or wiring	Check GPIO4 connection, verify GND
Expansion board pins confusing	Using ESP32 labels instead of board labels	Always follow expansion board labels - they're offset!
Can't upload code	Wrong boot mode	Hold BOOT, press RESET, release BOOT, then upload
ESP32 getting hot	Short circuit or wrong voltage	DISCONNECT IMMEDIATELY, check wiring
Lights cycle but no preemption	Button or GPS issue	Test button with Serial Monitor, check GPS data